

Setting Up GitHub

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Create a GitHub Account

- 1 Go to github.com → Sign up
- 2 Choose a [username](#) — pick something appropriate, you will share
- 3 Select the [Free](#) plan or the student's one [username@campus.unimib.it](#)
- 4 Verify your email address

Why Not Just Use a Password?

GitHub **discontinued password authentication** for Git operations in 2021.

Two alternatives are available:

SSH key

- Daily work from a fixed machine
- Set once, works forever
- Most comfortable for interactive use

Personal Access Token (PAT)

- Scripts and automation
- CI/CD pipelines
- Temporary or shared access

We will set up **SSH** — the recommended approach for interactive use.

What Is an SSH Key?

SSH (Secure Shell) authentication uses a **key pair**:

- 🔑 **Private key** — stays on your machine, **never shared**
- 👤 **Public key** — uploaded to GitHub, can be freely distributed

When you connect, GitHub challenges your machine with a message that only the private key can sign. No password is ever transmitted.

Analogy

The public key is like a padlock 👤 you give to GitHub. Only you hold the key 🔑 that opens it.

Step 1 — Check for Existing Keys

Open a terminal and run:

```
ls ~/.ssh
```

- If you see `id_ed25519` and `id_ed25519.pub` (or `id_rsa` / `id_rsa.pub`) — you already have a key pair

→ Skip to Step 3

- If the `.ssh` folder is empty or does not exist — generate a new key

→ Continue to Step 2

Step 2 — Generate a Key Pair

```
ssh-keygen -t ed25519 -C "your_email@example.com"
```

- Accept the default file location (press)
- Set a passphrase (recommended) or leave it empty

This creates two files in `~/.ssh/`:

`id_ed25519`

Your **private** key

Never share this file

`id_ed25519.pub`

Your **public** key

You will upload this to GitHub

Step 3 — Add the Key to the SSH Agent

The agent holds your key in memory so you do not need to type your passphrase at every push.

```
eval "$(ssh-agent -s)"  
ssh-add ~/.ssh/id_ed25519
```

Make it permanent

Add these two lines to your `~/.bashrc` (or `~/.zshrc`) so the agent starts automatically at login.

Step 4 — Copy Your Public Key

```
cat ~/.ssh/id_ed25519.pub
```

The output looks like:

```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAA... your_email@example.com
```

Select the entire line and copy to clipboard (**Ctrl** + **Shift** + **C** in the terminal).

Important

Copy the **.pub** file content — **not** the private key.

Step 5 — Add the Key to GitHub

- 1 GitHub → top-right avatar → [Settings](#)
- 2 Left sidebar → [SSH and GPG keys](#)
- 3 Click [New SSH key](#)
- 4 Title: something descriptive (e.g. Ubuntu VM Lab)
- 5 Paste your public key → [Add SSH key](#)

You can add multiple keys — one per machine you work on.

Step 6 — Test the Connection

```
ssh -T git@github.com
```

Expected response:

Hi username! You've successfully authenticated,
but GitHub does not provide shell access.

SSH setup is complete

From now on, any `git push` or `git pull` to GitHub will authenticate automatically.

Alternative: Personal Access Token (PAT)

Use a PAT when SSH is not practical (shared machine, automated script, CI/CD).

Generate a PAT:

- 1 GitHub → Settings → Developer settings
- 2 Personal access tokens → Tokens (classic) → Generate new token
- 3 Select scope: at minimum check repo
- 4 Copy the token immediately — GitHub will not show it again

Use the token as your password when Git asks for credentials over HTTPS.

Storing a PAT Safely

Never paste a token directly into a script or commit it to a repository.

Store it in your system credential manager.

```
git config --global credential.helper store
```

The first time you push, Git will ask for username + token and store it in `~/.git-credentials`.

For lab machines

Use cache instead of store so the token is only kept in memory for a limited time:

```
git config --global credential.helper 'cache --timeout=3600'
```

Setup Checklist

One-time setup

- ✓ Create GitHub account
- ✓ Generate SSH key pair
- ✓ Add public key to GitHub
- ✓ Test connection with `ssh -T`

Per-machine setup

- ✓ `git config --global user.name`
- ✓ `git config --global user.email`
- ✓ Start `ssh-agent` (or add to `~/.bashrc`)

Once done, you will never need to type a password for GitHub again.